Blockchain-Based Secure Communication System

1. Abstract

The Blockchain-Based Secure Communication System is a decentralized application designed to provide secure, tamper-proof, and private communication between users. Traditional communication systems rely on centralized servers, which are vulnerable to data breaches, censorship, and unauthorized access. This project leverages blockchain technology and IPFS (InterPlanetary File System) to ensure data integrity, confidentiality, and decentralization.

The system uses Ethereum blockchain for transaction verification and IPFS for storing encrypted messages. A React-based frontend enables user interaction, while a Node.js backend manages API requests and blockchain communication.

2. Introduction

2.1 Background

With increasing cyber threats, traditional communication platforms face challenges such as:

- Data interception
- Centralized vulnerabilities
- Lack of transparency

Blockchain provides:

- Decentralization
- Immutability
- Enhanced security

2.2 Objective

The main objectives of this project are:

- To develop a secure communication system using blockchain
- To eliminate dependency on centralized servers
- To ensure data integrity and confidentiality
- To integrate IPFS for decentralized storage.

3. System Architecture

The system consists of three main components:

3.1 Frontend

- Built using React.js
- Provides user interface for sending and receiving messages
- Connects with blockchain using Ethers.js

3.2 Backend

- Developed using Node.js
- Handles API requests
- Manages blockchain interactions

3.3 Blockchain & Storage

- Ethereum Blockchain: Stores transaction hashes
- IPFS: Stores encrypted messages

4. Technologies Used

Technology	Purpose
React.js	Frontend UI
Node.js	Backend server
Ethers.js	Blockchain interaction
IPFS	Decentralized storage
Ethereum	Smart contract platform
Vite	Frontend build tool

5. Working of the System

Step 1: Message Creation

- User writes a message in the frontend
- Message is encrypted

Step 2: Upload to IPFS

- Encrypted message is uploaded to IPFS
- IPFS returns a unique hash (CID)

Step 3: Blockchain Transaction

- The IPFS hash is stored on the blockchain
- Transaction ensures immutability

Step 4: Message Retrieval

- Receiver fetches hash from blockchain
- Retrieves message from IPFS
- Decrypts the message

6. Features

- End-to-end encryption
- Decentralized storage (IPFS)
- Blockchain-based verification
- Secure identity handling
- Tamper-proof communication

7. Advantages

- No central authority
- High security and privacy
- Data cannot be altered once stored
- Resistant to cyber attacks

8. Limitations

- Transaction cost (Gas fees)
- Slower compared to centralized systems
- Requires blockchain knowledge for deployment

9. Future Enhancements

- Integration with mobile applications
- Use of Layer-2 solutions to reduce gas fees
- Advanced encryption techniques
- Real-time messaging with WebSockets
- User authentication with biometric security

10. Applications

- Secure government communication
- Military messaging systems
- Corporate confidential communication
- Healthcare data exchange

11. Conclusion

The Blockchain-Based Secure Communication System demonstrates how decentralized technologies can enhance privacy, security, and trust in communication systems. By combining blockchain with IPFS, the project successfully eliminates reliance on centralized servers and ensures data integrity. This system can be further scaled and optimized for real-world applications.

12. References

1. Ethereum Documentation
2. IPFS Documentation
3. Ethers.js Library
4. React.js Documentation
5. Node.js Documentation

13. Appendix (Project Structure)

```
blockchain_comm_sys/

│

├── backend/

│   ├── index.js

│   └── package.json

│

├── frontend/

│   ├── index.html
```

```
│   ├── package.json
│   └── node_modules/
│
├── package.json
├── package-lock.json
└── README.md
```

14. How to Run the Project

Backend

cd backend

npm install

node index.js

Frontend

cd frontend

npm install

npm run dev

15. Result

The system successfully:

- Stores messages securely on IPFS
- Records message references on blockchain
- Allows secure retrieval and decryption

16. Acknowledgment

We would like to express our gratitude to our mentors and resources that guided us throughout the development of this project.